

Student Enrollment & Course Management System (SECMS)

A Salesforce Lightning Application

Project Report

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Platform: Salesforce Developer Edition

1. Introduction

1.1 Project Overview

The Student Enrollment & Course Management System (SECMS) is a Salesforce application that centralises student, course, instructor and enrollment data in one place. Enrollment Officers manage admissions and registrations, Administrators monitor revenue and approvals, and Instructors see the enrollments assigned to them. The system uses Salesforce Flows, Approval Processes, Validation Rules, Reports and Dashboards to automate the work from admission to course completion.

1.2 Purpose

SECMS replaces manual and scattered enrollment handling with a secure, automated Salesforce solution that provides:

- Accurate student data management
- Structured enrollment workflows
- Automated approvals and validations
- Transparent reporting and monitoring

2. Ideation Phase

2.1 Problem Statement

	I am	I am trying to	But	Because	Which makes me feel
PS-1	Enrollment Officer	manage student enrollments efficiently	enrollment data is scattered	systems are manual and unintegrated	frustrated and error-prone
PS-2	Administrator	track enrollments and revenue	reporting is delayed	data is not centralised	uncertain in decision-making

2.2 Empathy Map (Enrollment Officer)

- Think & Feel: Concerned about double bookings and revenue loss
- Hear: Complaints from students and management
- See: Multiple spreadsheets and overflowing emails
- Say & Do: Rechecks data, follows up manually
- Pain: Duplicate entries, approval bottlenecks
- Gain: Automation, dashboards, faster processing

2.3 Brainstorming & Idea Prioritization

Ideas considered: spreadsheet automation, standalone enrollment portal, ERP integration and a Salesforce-based enrollment system. They were grouped as manual, semi-automated and fully automated cloud solutions. The Salesforce-based SECMS was selected because it has both high feasibility and high impact.

3. Requirement Analysis

3.1 Customer Journey

Awareness & Registration → Course Selection → Enrollment Request → Fee Payment → Approval → Instructor Assignment → Reporting & Completion.

3.2 Functional Requirements

FR No	Epic	Description
FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve and reject enrollments
FR-4	Instructor Assignment	Auto-assign instructors by course category
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR	Requirement	Example
NFR-1 Usability	Simple, intuitive interface needing minimal training	Role-based app, tabs, one-click approval
NFR-2 Security	Only authorised users access sensitive data	Instructors see only assigned enrollments (Enrollment OWD = Private)
NFR-3 Reliability	Automation runs correctly without duplicates or wrong status	Fees must be paid before approval
NFR-4 Performance	Fast response, especially in admission season	Quick dashboard refresh, fast search
NFR-5 Scalability	Supports more students, courses and departments	Add departments without redesign
NFR-6 Availability	Accessible during working hours with minimal downtime	24/7 cloud access, Salesforce backups

3.3 Data Flow

The user enters student and enrollment data. The system validates and stores the records in Salesforce objects. The approval process routes the enrollment to the Training Manager, flows assign the instructor and update statuses, and reports and dashboards consume the processed data.

3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

4. Project Design

4.1 Problem–Solution Fit

Fragmented spreadsheets, manual paperwork, delayed approvals and scattered systems are replaced by unified dashboards, automated approvals and a centralised database on the Salesforce platform.

4.2 Proposed Solution

Parameter	Description
Problem	Manual enrollment handling
Solution	Salesforce-based SECMS
Novelty	End-to-end automation
Social Impact	Improved student experience
Revenue Model	Institutional deployment
Scalability	Multi-department support

4.3 Solution Architecture

Four custom objects (Student, Course, Instructor, Enrollment) sit on the Salesforce Platform. Enrollment links to the other three through lookup relationships. Automation flows, approval processes, dashboards and the security layer work on top of these objects.

Data Model

Object	Key Fields
Student__c	Student Name, Email, Phone, Date of Birth, Status (picklist), Address
Course__c	Course Name, Category (Technical / Language / Non-Technical), Duration, Course Fee, Description
Instructor__c	Instructor Name, Instructor Code (INSTR_A/B/C), Expertise, Phone, Email
Enrollment__c	Student (Lookup), Course (Lookup), Instructor (Lookup), Enrollment Date, Enrollment Status (Requested / Approved / Rejected), Fees Paid (Checkbox), Total Amount, Comments

5. Project Planning & Scheduling

Agile methodology was followed: Epics → User Stories → Story Points, executed in six-day sprints, with velocity used for estimation.

5.1 Product Backlog

Sprint	Epic	USN	User Story / Task	Pts	Priority	Member
1	Developer Setup	USN-1	Create a Salesforce developer account to configure and deploy the system	1	High	Member 1
1	Data Modeling	USN-2	Create custom objects Student, Course, Enrollment, Instructor	3	High	Member 2
2	Data Modeling	USN-3	Dedicated tabs for each object	3	High	Member 3
2	Data Modeling	USN-4	Create fields for student, course and enrollment details	5	High	Member 4
2	Data Modeling	USN-5	Centralised Lightning App for all operations	3	High	Member 4
2	Data Modeling	USN-6	Define relationships between students, courses and enrollments	5	Medium	Member 1
2	Data Modeling	USN-7	Customised page layouts and record types	5	High	Member 2

Sprint	Epic	USN	User Story / Task	Pts	Priority	Member
3	Automation	USN-8	Validation rules for mandatory data and fee payment	3	High	Member 4
3	Automation	USN-9	Flows for enrollment processing, notifications and instructor assignment	3	High	Member 4
3	Automation	USN-10	Approval process for enrollment requests	5	High	Member 1
4	Security	USN-11	Role-based security (profiles, permission sets, OWD)	5	High	Member 2
5	Reports	USN-12	Reports on enrollments, courses and revenue	4	High	Member 3
6	Dashboards	USN-13	Visual dashboard for enrollment statistics	4	Medium	Member 4

5.2 Sprint Schedule, Velocity & Burndown

Sprint	Total Points	Duration	Start	Planned End	Completed	Release
Sprint-1	4	6 days	03 Nov 2025	08 Nov 2025	4	08 Nov 2025
Sprint-2	21	6 days	10 Nov 2025	15 Nov 2025	21	15 Nov 2025
Sprint-3	11	6 days	17 Nov 2025	22 Nov 2025	11	22 Nov 2025
Sprint-4	5	6 days	24 Nov 2025	29 Nov 2025	5	29 Nov 2025
Sprint-5	4	6 days	01 Dec 2025	06 Dec 2025	4	06 Dec 2025
Sprint-6	4	6 days	08 Dec 2025	13 Dec 2025	4	13 Dec 2025

Total story points = 49 across 6 sprints. Average velocity = $49 / 6 \approx 8.2$ points per sprint (6-day sprint, so ≈ 1.36 points per day). Remaining points after each sprint: 45, 24, 13, 8, 4, 0, which forms the burndown line.

6. Project Development

The project was built in the following milestones in a Salesforce Developer Edition org.

No.	Milestone	What was done
1	Developer account	Sign up at developer.salesforce.com/signup , verify email, set password.
2	Objects	Created Student, Course, Instructor and Enrollment custom objects with Reports, Activities, Field History and Search enabled.
3	Fields	Added fields listed in the Data Model (picklists for Status, Category, Expertise, Enrollment Status).
4	Relationships	Three lookups from Enrollment: to Student, Course and Instructor.
5	Validation Rules	(a) Student email must contain @student.edu. (b) Enrollment cannot be Approved unless Fees Paid is checked.
6	Approval Process	Approval_Enrollment_Request: entry criteria Status = Requested; approver Training Manager; approved → Status = Approved; rejected → Status = Rejected and rejection email alert.
7	Tabs	Custom tabs for all four objects.
8	Lightning App	Student Enrollment & Course Management app with object tabs, Reports, Dashboards, Home and Tasks.

No.	Milestone	What was done
9	Record Types & Layouts	New Enrollment and Re-Enrollment record types with separate layouts; Enrollments related list added to Student, Course and Instructor.
10	Flows	SetEnrollmentDate, Enrollment Approved Action, AutoAssign_Instructor, Create_Followup_Task.
11	Reports	Students by Status, Enrollments by Course, Pending Enrollment Approvals, Revenue Report – Fees Paid.
12	Dashboards	Student Management Dashboard with donut/bar, bar, gauge and table components.
13	Security	Profiles (Admin, Enrollment Officer, Instructor), permission sets, and OWD.

6.1 Flow Details

Flow	Trigger	Logic
SetEnrollmentDate	Enrollment created	Update Records: Enrollment_Date__c = \$Flow.CurrentDate
Enrollment Approved Action	Enrollment created/updated, Status = Approved	Get Records: Student where Id = Enrollment.Student__c; Email Alert to student; Update Records: Student.Status__c = Active
AutoAssign_Instructor	Enrollment created/updated	Get Course; Decision on Category: Technical → Instructor A, Language → Instructor B, Non-Technical → Instructor C; Update Enrollment.Instructor__c
Create_Followup_Task	Enrollment Status = Requested	Get Admin user; formula TODAY()+2; Create Task "Follow up on enrollment request" related to the Enrollment

6.2 Security Setup

Item	Configuration
Admin profile	Full access
Enrollment Officer	Enrollment: Read, Create, Edit; Student and Course: Read (Edit if required)
Instructor	Read-only on assigned Enrollments
Permission sets	Custom Object Access; Dashboard Editor
OWD	Student: Public Read/Write; Course: Public Read/Write; Enrollment: Private

7. Functional & Performance Testing

Each feature was tested in the Salesforce org. Screenshots of each test are attached in the Appendix.

ID	Test Scenario	Expected Result	Actual Result	Status
TC-01	Create Student with email ending @student.edu	Record saved	Saved successfully	Pass
TC-02	Create Student with email gmail.com	Error: Email must be a valid student institutional email	Error shown on Email field	Pass
TC-03	Create Course with Category and Fee	Record saved	Saved successfully	Pass
TC-04	Create Instructor with code INSTR_A	Record saved	Saved successfully	Pass

ID	Test Scenario	Expected Result	Actual Result	Status
TC-05	Create Enrollment (Student, Course, Instructor, Status = Requested)	Record saved; Enrollment Date auto-filled with today	Date auto-filled	Pass
TC-06	Set Status = Approved with Fees Paid unchecked	Error: Fees must be paid before approving an enrollment	Error shown at top of page	Pass
TC-07	Set Status = Approved with Fees Paid checked	Saved; student email sent; Student Status = Active	Updated as expected	Pass
TC-08	Enrollment with Technical course	Instructor A auto-assigned	Instructor A assigned	Pass
TC-09	Enrollment with Language course	Instructor B auto-assigned	Instructor B assigned	Pass
TC-10	Enrollment with Requested status	Follow-up task created for Admin, due in 2 days	Task created	Pass
TC-11	Submit enrollment for approval; Training Manager rejects	Status = Rejected; rejection email sent	Status updated	Pass
TC-12	Run all four reports	Correct grouping and counts	Matches records	Pass
TC-13	Open Student Management Dashboard	All components show data	Dashboard displayed	Pass
TC-14	Login as Instructor	Only assigned enrollments visible (Enrollment OWD Private)	As configured	Pass

Performance: record creation, flow execution, report and dashboard refresh completed within a few seconds on the sample data set (about 5 students, 3 courses, 3 instructors and 6 enrollments).

8. Advantages & Disadvantages

Advantages

- Centralised data
- Automated workflows
- Real-time reporting
- Role-based security

Disadvantages

- Salesforce dependency (licence and platform limits)
- Requires initial configuration effort

9. Conclusion

SECMS streamlines the student academic lifecycle: registration, course creation, enrollment management, instructor assignment, approvals, revenue tracking and reporting. Using validation rules, flows, approval processes and dashboards, the system improves data accuracy, efficiency and decision-making. The project shows how Salesforce can turn manual academic processes into a scalable, reliable and user-friendly digital solution.

10. Future Scope

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

11. Appendix

- Source: Salesforce configuration (declarative, no Apex code)
- Dataset: Salesforce custom objects (sample data below)
- Demo: Salesforce Developer Org

Sample Data

Instructors

Name	Code	Expertise	Email
Instructor A	INSTR_A	Technical	instra@example.com
Instructor B	INSTR_B	Language	instrb@example.com
Instructor C	INSTR_C	Non-Technical	instrc@example.com

Courses

Course	Category	Duration	Fee
Computer Science	Technical	6 months	25000
Electronics & Communication Engineering	Technical	6 months	22000
English Communication	Language	3 months	8000
Business Management	Non-Technical	4 months	12000

Students

Name	Email	Phone	Status
Abhishek Jakkula	abhishek@student.edu	9000000001	Active
Vignesh Doodh	vignesh@student.edu	9000000002	Active
Swaroop M	swaroop@student.edu	9000000003	Active
Sushanth Jaligama	sushanth@student.edu	9000000004	Active

Enrollments

Student	Course	Status	Fees Paid
Sushanth Jaligama	Computer Science	Requested	No
Vignesh Doodh	Electronics & Communication Engineering	Requested	No
Swaroop M	English Communication	Approved	Yes
Abhishek Jakkula	Business Management	Rejected	No

Screenshots to attach: objects and fields, relationships, both validation errors, approval process, each flow canvas, test records, four reports, dashboard, profiles and OWD. (Hide any personal email addresses before pasting.)

Thank You